[INCH-POUND]
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20 January 1984

COMMERCIAL ITEM DESCRIPTION

BLOCKS, TACKLE, WIRE ROPE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers single-and multiple-sheave, oval pattern, diamond pattern, snatch, and cargo-hoisting blocks used with wire rope for use in various rigging, lifting, towing, and construction operations.

2. CLASSIFICATION. The blocks shall be of the following types, classes, styles, and sizes. The size of the block is determined by the outside diameter of the sheave. See table I for block size and availability of types and classes. The type, class, style, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Oval pattern
Type II - Diamond pattern
Type III - Snatch
Type IV - Cargo hoisting

Class 1 - Regular duty
Class 2 - Heavy duty
Class 3 - Extra heavy duty

Style A - Single sheave
Style B - Double sheave
Style C - Triple sheave

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3940

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TABLE I. Block sizes.

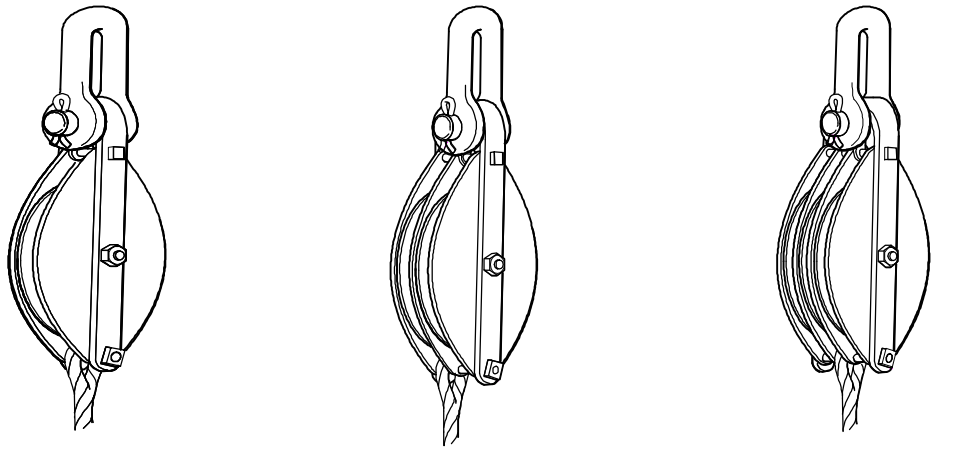
Size	Sheave diameter, inches	Types I and II (styles A, B, and C)			Type III (styles A and B)			Type IV (style A)	
		Class 1	Class 2	Class 3	Class 1	Class 2	Class 3	Class 1	Class 2
4	4	X			X	X			
6	6	X	X	X	X	X	X	X	X
8	8	X	X	X	X	X	X	X	X
10	10	X	X	X	X	X	X	X	X
12	12	X	X	X	X	X	X	X	X
14	14	X		X	X	X	X	X	X
16	16	X		X	X	X	X	X	
17	17			X					
18	18			X		X		X	
20	20			X	X	X			
24	24			X		X			

Note: "X" denotes availability of block sizes. Type III blocks are not available in style C. Type IV blocks are not available in class 3 and styles B and C.

3. SALIENT CHARACTERISTICS

3.1 General requirements. Block components shall be fabricated from compatible materials and shall be inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration during use and storage. All parts subject to wear, breakage, or distortion shall be accessible for inspection, adjustment, replacement, and repair.

3.1.1 Type I blocks. Type I blocks shall have oval pattern shells as shown in figure 1.

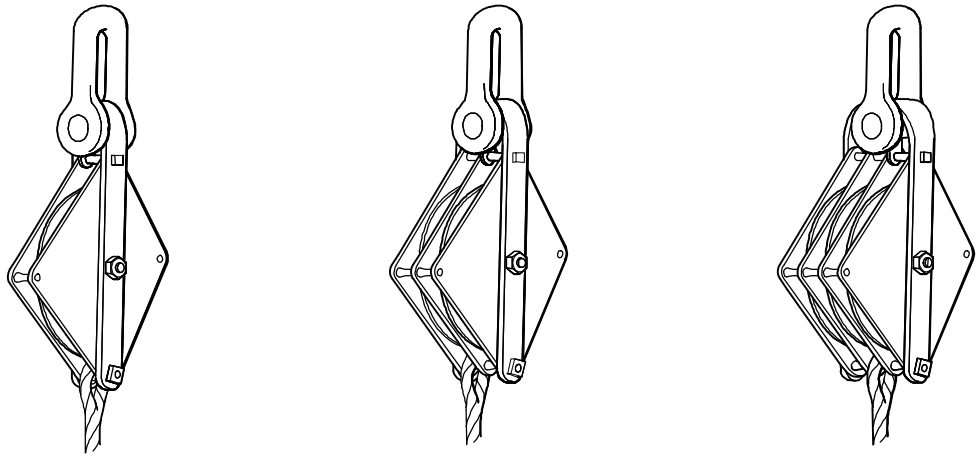


Style A - Single Style B - Double Style C - Triple

FIGURE 1. Type I oval pattern blocks.

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3.1.2 Type II blocks. Type II blocks shall have diamond pattern shells with guide rollers front and back as shown in figure 2.



Style A - Single Style B - Double Style C - Triple

FIGURE 2. Type II diamond pattern blocks.

3.1.3 Type III blocks. Type III blocks shall have connections and fittings that swing out of position to permit quick entry of the wire rope into sheave grooves without reeving. When specified (see 7.2(c)), type III blocks shall have a self-locking mechanism to keep the block securely closed while under load. Figure 3 shows a typical type III, style A block.

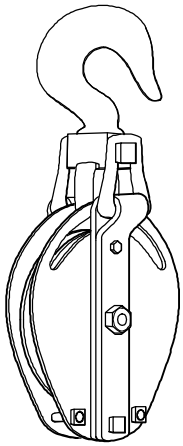


FIGURE 3. Type III snatch block.

3.1.4 Type IV blocks. Type IV blocks shall be half-side or full-side blocks as specified (see 7.2(d)). When specified (see 7.2(e)), the shell shall be fixed to the shackle with two removable draw pins chained together. Figures 4 and 5 show typical configurations for type IV, style A blocks.

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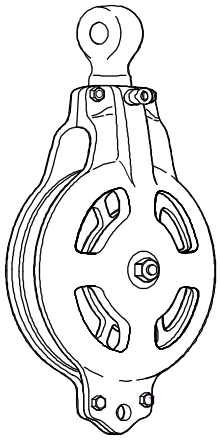


FIGURE 4. Cargo hoisting block.

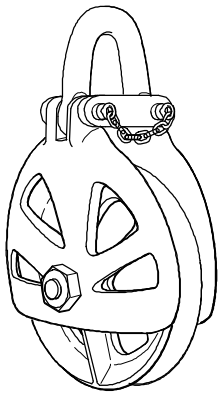


FIGURE 5. Cargo-hoisting block with removable chained pins.

3.2 Components. The blocks shall include, as a minimum, the following components:

3.2.1 Shell. Unless otherwise specified (see 7.2(f)), shell plates shall be made of cast steel, structural steel, or carbon steel hot-rolled sheet conforming to American Society for Testing and Materials (ASTM) standard A27, "Standard Specification for Steel Castings, Carbon, for General Application", standard A36, "Standard Specification for Carbon Structural Steel", or standard A569, "Standard Specification for Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip Commercial", respectively. Material selection and heat treatment shall be as required to meet the proof and ultimate strength test conditions. The edges shall be well rounded.

3.2.2 Strap. Unless otherwise specified (see 7.2(g)), steel straps shall be made of structural steel, carbon steel cold-rolled sheet, or carbon steel hot-rolled sheet conforming to ASTM standard A36, "Standard Specification for Carbon Structural Steel", standard A109, "Standard Specification for Steel, Strip, Carbon, Cold-Rolled", or standard A569, "Standard Specification for Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip Commercial", respectively. Heat treatment shall be as required to meet the proof and ultimate strength test conditions. Steel straps shall support the center pin and transmit sheave loads to the fitting (e.g., shackle, hook, eye) as required.

3.2.3 Sheave(s). Unless otherwise specified (see 7.2(h)), the sheaves(s) shall be made of cast steel conforming to ASTM standard A27, "Standard Specification for Steel Castings, Carbon, for General Application", or forged steel conforming to ASTM standard A668, "Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use". The sheave(s) shall be designed to prevent cutting or other damage to the rope and shall have smooth-running action. The rim and tread section of each sheave shall be properly grooved to fit the contour of the rope under load conditions. The sheave(s) shall be concentric with the bore and shall be designed so that only the hub bears against the shell. Side play shall not exceed 1/8 inch.

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3.2.4 Bearing(s). Bearings shall be either self-lubricating bronze bushing or pressure-lubricated bronze bushing, roller, tapered roller, or ball bearing as specified (see 7.2(i)). Self-lubricating bushings shall conform to ASTM standard B438, "Standard Specification for Sintered Bronze Bearings (Oil-Impregnated)". Grease fittings for pressure-lubricated bearings shall be recessed and shall conform to Society of Automotive Engineers (SAE) standard J534, "Lubrication Fittings".

3.2.5 Center pin. Unless otherwise specified (see 7.2(j)), the center pin shall be of carbon steel conforming to ASTM standard A108, "Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality". The pin shall transmit the sheave load to the shell or straps without rotating. Heat treatment shall be as required to meet proof and ultimate strengths. The pin may be drilled to accommodate the pressure lubrication system.

3.2.6 Fittings. If required, each block shall include a fitting (e.g., shackle, hook, eye) as specified (see 7.2(k)).

3.2.7 Becket. When specified (see 7.2(l)), the becket shall be formed by the extension of the strap ends to provide a suspension point for a wire rope thimble.

3.2.8 Spacers and rollers. Spacers shall be provided as required to make the block a strong, rugged unit with the required sheave clearances. When specified (see 7.2m()), rollers or guides shall be provided to prevent the wire from coming out of the sheaves when the block is overhauled.

3.2.9 Optional components. Optional components shall be provided as specified (see 7.2(n)).

3.3 Performance. Unless otherwise specified (see 7.2(o)), the block shall meet the performance requirements shown in tables II through IX when tested in accordance with sections 5.3 through 5.3.3.

TABLE II. Type I and II, class 1 block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds		
		Style A	Style B	Style C
4	1/4	2,000	3,000	4,000
6	3/8	4,000	5,000	6,000
8	1/2	5,000	7,000	9,000
10	5/8	6,000	9,000	11,000
12	3/4	8,000	12,000	14,000
14	3/4	10,000	14,000	16,000
16	7/8	16,000	30,000	36,000

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TABLE III. Type I and II, class 2 block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds		
		Style A	Style B	Style C
6	3/8	5,000	6,000	8,000
8	1/2	7,000	12,000	14,000
10	5/8	8,000	14,000	16,000
12	5/8	14,000	24,000	28,000

TABLE IV. Type I and II, class 3 block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds		
		Style A	Style B	Style C
6	1/2	15,000	20,000	25,000
8	3/4	15,000	20,000	25,000
10	3/4	15,000	20,000	25,000
12	3/4	20,000	30,000	40,000
14	3/4	20,000	40,000	60,000
16	7/8	30,000	50,000	70,000
17	7/8	30,000	50,000	70,000
18	1	36,000	60,000	90,000
20	1	40,000	70,000	90,000
24	1-1/8	50,000	80,000	100,000

TABLE V. Type III, class 1 block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds	
		Style A	Style B
04	3/8	2,000	2,000
06	3/8	4,000	4,000
08	1/2	6,000	6,000
10	5/8	8,000	8,000
12	3/4	10,000	10,000
14	3/4	13,000	13,000
16	7/8	15,000	15,000
20	1-1/4	24,000	24,000

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TABLE VI. Type III, class 2 block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds	
		Style A	Style B
04	1/4	3,500	
06	5/8	7,000	25,000
08	5/8	10,500	25,000
10	5/8	16,000	25,000
12	3/4	23,000	30,000
14	7/8	31,000	44,800
16	7/8	33,000	
18	7/8	33,000	
20	1-1/8	65,000	
24	1-1/8	65,000	

TABLE VII. Type III, class 3 block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds	
		Style A	Style B
6	5/8	10,000	10,000
8	3/4	20,000	20,000
10	7/8	30,000	30,000
12	1	50,000	50,000
14	1-1/8	80,000	80,000
16	1-1/4	90,000	90,000

TABLE VIII. Type IV, class 1, style A block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds
6	3/8	6,500
8	3/8	8,500
10	1/2	10,000
12	3/4	14,600
14	3/4	20,000
16	3/4	20,000
18	3/4	20,000

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TABLE IX. Type IV, class 2, style A block safe working loads.

Size (sheave diameter, inches)	Wire rope diameter, inches	Safe working load (SWL), pounds
6	3/8	12,500
8	1/2	16,000
10	1/2	16,000
12	1/2	27,000
14	3/4	35,000

3.4 Safety and health requirements. The manufacturers shall ensure that the blocks and all components and fittings used on the blocks shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR Part 1910.184, "Slings". If a conflict arises between the specifications in the CID and the OSHA standards, the OSHA standards shall apply.

3.5 Identification markings. Unless otherwise specified (see 7.2(p)), the blocks shall be permanently marked with the following information:

a. Manufacturer's name or trademark
b. Manufacturer's part number
c. Size of wire rope
d. Safe working load (SWL)
e. Proof load

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; shall conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and shall be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The block shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(q)), the block shall be tested in accordance with sections 5.3.1 through 5.3.3.

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5.3.1 Operational test. The block shall be operated in accordance with the manufacturer's standard operating procedures. The block shall be operated to ensure that all components operate freely without restriction or binding.

5.3.2 Proof load test. Each block shall be proof tested to twice the SWL shown in tables II through IX for the block selected. The applicable proof test load shall be applied to the block assembly and held for a period of not less than 10 minutes. At the conclusion of the test, the block and its components shall be carefully examined and measured for signs of fracture, permanent set, or evidence of instability, deformation, excessive wear, or mechanical defects. When a becket is provided, it shall show no distortion with the wire rope dead-ended at the becket and twice the SWL shown in tables II through IX applied to the block selected.

5.3.3 Ultimate strength test. The manufacturer shall certify that when tested to destruction, the block did not fail at loads less than 5 times the SWL shown in tables II through IX for the block selected. Failure of the fittings (e.g., shackle, hook, or eye) shall occur before failure of any other components of the block and must show a clear visual indication of deformation before failure.

5.4 Acceptance. Unless otherwise specified (see 7.2(r)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the performance requirements of section 3.3. Failure of the block to meet the performance requirements of section 3.3 shall be cause for rejection. The manufacturer shall provide certification with each block.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products supplied shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(s)).

7. NOTES

7.1 Sources of documents

7.1.1 ASTM standards. Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.1.2 SAE standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.

7.1.3 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.1.4 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

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7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Type, class, style, and size of block required (see 2)
c. Type III block with self-locking mechanism, if required (see 3.1.3)
d. Type IV block with half side or full side, as specified (see 3.1.4)
e. Type IV block with removable pins, if required (see 3.1.4)
f. Shell material, if different (see 3.2.1)
g. Strap material, if different (see 3.2.2)
h. Sheave material, if different (see 3.2.3)
i. Bearing type and lubrication method, as specified (see 3.2.4)
j. Center pin material, if different (see 3.2.5)
k. Fitting type, as specified (see 3.2.6)
l. Becket, if required (see 3.2.7)
m. Guides and rollers, if required (see 3.2.8)
n. Optional components, as specified (see 3.2.9)
o. Performance, if different (see 3.3)
p. Identification marking, if different (3.5)
q. Testing, if different (see 5.3)
r. Acceptance, if different (see 5.4)
s. Packaging requirements, (see 6.1)

7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.

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- Size: (see table I)
- Style: A - Style A - Single sheave; B - Style B - Double sheave; C - Style C - Triple sheave
- Class: 1 - Class 1 - Regular duty; 2 - Class 2 - Heavy duty; 3 - Class 3 - Extra heavy duty
- Type: A - Type I - Oval pattern; B - Type II - Diamond pattern; C - Type III - Snatch; D - Type IV - Cargo hoisting
- CID number

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MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AT
Navy - SH

Reviewers:
Air Force - 84
Army - CR
Navy - MC, YD2

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS

(Project 3940-0017)

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